Master Thesis — Progress Report
Spain's MTU15 reform: empirical assessment in progress

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Outline

- **1. Reminder.** Objective, research questions, context.
- **2. What I have done.** Data, methods, headline findings, tests of theory.
- **3. State of the work.** Claim portfolio, theoretical model.
- **4. Difficulties encountered.** Identification, an unexplained break.
- **5. What is left.**

Following advisor's guidance: a progress report rather than a results talk. 18–20 minutes.

Objective and concrete research questions

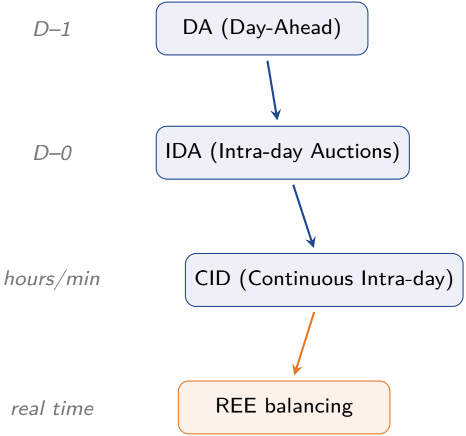
Aim

Empirically assess the impact of Spain's 2024–2025 **MTU15 reform sequence** on market outcomes and firm strategic behaviour, and extend the theoretical framework to capture the mechanism.

Five concrete sub-questions:

1. Does the **asymmetric-clock window** (DA60 + ID15 + ISP15) generate a measurable dual-pricing redistribution from BRPs to the TSO? Does it close at MTU15-DA?
2. Does finer market-time-unit granularity **reduce strategic forward-quantity withholding** (Allaz–Vila / Ito–Reguant prediction)?
3. Does the reform create **heterogeneous winners and losers** across firms and technologies?
4. Does the per-segment incidence of imbalance settlement match a **Pigouvian benchmark**?
5. Has *operación reforzada* (post-blackout regime) broken the separability of the reform sequence?

Big picture — D–1 to delivery



- **DA** (uniform-price auction): main stage, sets baseline schedule and price.
- **IDA** (uniform-price auctions): three discrete sessions to update positions as forecasts and availability change.
- **CID** (continuous matching, XBID/SIDC): pay-as-bid, last voluntary adjustment.
- **REE balancing** + **imbalance settlement**: residual deviations + system security; ex-post dual-pricing rule (EBGL Art. 52).

The 2024–2025 reform sequence

- Traditionally MTU was **60 min** everywhere.
- Three stages:
  - **Dec 2024 — ISP15**: imbalance settlement → 15-min.
  - **Mar 2025 — ID15**: intraday → 15-min.
  - **Oct 2025 — DA15**: day-ahead → 15-min.
- Two transitional regimes where *settlement* and *trading* clocks differ.
- Plus: post-2025-04-28 blackout ⇒ *operación reforzada* (REE's reinforced-operation regime), continuous since.

Period	ISP	DA	ID
Pre-Dec 2024	60	60	60
Dec24–Mar25	**15**	60	60
Mar25–Sep25	15	60	**15**
Oct25 onward	15	**15**	15

ISP15 | ID15 | DA15

Dec 2024 | *Mar 2025* | *Oct 2025*

Data pipeline

Three sources, $\sim$8 years (2018-01 $\rightarrow$ 2026-02), all idempotent and version-controlled.

OMIE (Iberian market operator) — the primary source

25+ data families at native MTU60/MTU15 granularity:

- *Prices*: day-ahead, intraday auctions, continuous market.
- *Programs*: PDBC (DA cleared), PDBF (+ bilaterals, 166M rows), per-firm/per-unit, intraday, final hourly.
- *Bid curves*: DA + IDA offer headers + tranches; XBID limit orders + matched trades.
- Plus interconnection capacities and annulments.

ENTSO-E Transparency Platform (pan-European)

Spanish control area, API-pull. DA/imbalance prices, FRR settlement (A87), generation by type and unit, load actual + forecast, wind+solar actual + forecasts, reservoir filling. *All fundamentals controls and forecast errors come from here.*

REE / ESIOS

Settlement detail (per-firm, per-segment), real-time RT prices, balancing offers.
Caveat: ESIOS API access still pending — 3-week wait. Public archive endpoints work; per-BRP feeds blocked until the API token clears.

Empirical strategy

- **No clean external counterfactual** — most comparable EU markets adopted MTU15 already and differ structurally.
- Reduced-form before/after with rich controls:
  - Fundamentals: load, wind + solar (actual + forecast errors), fuel/CO_2.
  - Calendar: cal-month FE, day-of-week, hour-of-day; year FE where appropriate.
  - Reform-date and *reforzada* indicators.
- **Same-calendar-month robustness** mandatory for any cross-regime claim (Spain is highly seasonal).
- **Cluster-robust SE** by date or week-of-sample (Cameron–Miller $G \geq 30$ rule).
- **Cross-country placebo**: France DA prices flat across Spanish reform dates.
- **Structural-time-series counterfactual** (Brodersen et al. 2015): ongoing.

Headline findings

1. System-layer settlement transfer

During the asymmetric-granularity window (Dec 2024–Sep 2025) the BRP$\rightarrow$TSO net transfer was **+€1,094.9M cumulative** vs same-calendar pre-reform baseline (bootstrap CI $[-90, +73]$M $\Rightarrow \approx 15\times$ upper bound). Collapses to **€7.4M/mo at MTU15-DA** ($12\times$ drop). Regulatory *redistribution*, not deadweight.

2. Mechanism — forecast-error pass-through to imbalance

Pooled regression, daily grain: net per-regime slope β is ~ 0 pre-reform, ~ 0 at 3-session IDA, -0.03 at ISP15-win, $+0.09$ **at DA60/ID15** (uniquely elevated), ~ 0 at DA15/ID15. Robust to year FE.

3. Firm-level market power (Iberdrola)

Synthetic-firm replication *à la* Ciarreta–Espinosa: **Iberdrola alone** $\sim$**€820M** day-ahead cleared-price-difference rent post-MTU15-IDA (98% of joint Big-4); hydro 64% / CCGT 36%. **Regime-invariant.** Behavioural complement: Big-4 voluntary IDA repositioning (q_2, à la Ito–Reguant) collapses progressively, recovers at DA15/ID15; robust under canonical Fabra–Imelda controls.

Tests of theoretical predictions — with the literature

What worked:

- **Ito & Reguant (2016, AER)** sequential markets, market power, arbitrage: predicted q_2 collapse if firms react to a finer spot grid $\rightarrow$ *confirmed* (regime coefficients vary $< 0.1\%$ across spec battery).
- **Ciarreta & Espinosa (2010, J Regul Econ)** synthetic-firm method $\rightarrow$ *replicated and extended* (€820M IB rent; survives sensitivity tests).

What did not work:

- **Allaz & Vila (1993, JET)** forward markets dispel rents: my test was the slope $\partial \Delta Q_{IDA} / \partial q_{DA}$ $\rightarrow$ *rejected as a test* ($Q_{actual} \approx q_{DA} + \Delta Q_{IDA}$ is a mechanical accounting identity, not a structural slope). The surviving Allaz–Vila evidence comes from the Ito–Reguant q_2 trajectory above.
- **Hortaçsu & Puller (2008, RAND)** sophistication: implied Cournot–FOC Lerner vs realised marginal-bid markup $\rightarrow$ *strategic-conduct interpretation rejected* (implied ≈ 0.93 vs realised ≈ 0.34 for the leading firm; implied is formula-mechanical).
- **Bushnell (2003, JIE)** hydro market power via reservoir shadow-price $\rightarrow$ *mechanism rejected* (no reservoir gradient; placebo stronger than treatment). The IB hydro Q4 concentration is real but not classical Bushnell.

State of the work

Empirical portfolio:

- I track findings formally with a status (alive, wounded, dead) and a documented robustness trail.
- Currently: ∼40 alive, ∼7 wounded, ∼20 dead (audit trail).
- Each cited number is regression-backed at proper grain and survives multi-spec robustness.

Three policy levers identified:

1. Clock-symmetry — *implemented* at MTU15-DA.
2. Pigouvian incidence — *open*.
3. Operational regime overlay (*operación reforzada*) — open.

Theoretical model:

- Started from an extension of Ito–Reguant (2016) with explicit MTU mismatch (in my Energy Economics proposal).
- Empirically the three predictions hold; magnitudes are larger than the theory suggested.
- **Pivot**: *very solid empirics* + a *simplified IO model* for mechanism + a light first-order-condition exercise.
- **Not** a fully estimated structural auction model.

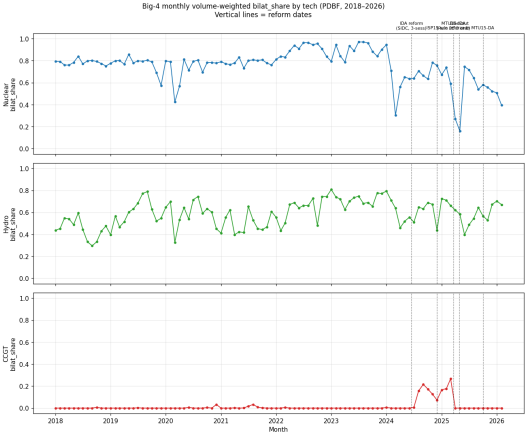
Difficulty 1 — identification under multiple confounds

- Each reform date has at least one competing event:
  - *Dec 2024*: European gas-price normalisation.
  - *Mar 2025*: CNMC eliminates the bilateral-contract opportunity-cost rule simultaneously.
  - *Apr 28 2025* blackout → *operación reforzada* (regime overlay continuous since).
  - *Oct 2025*: partial overlap with reforzada.
- **Strategy adopted:**
  - *Market-chain placement*: reforzada acts *after* the day-ahead clearing ⇒ DA-side outcomes face it only via expectations.
  - *Reforzada-held-constant pairings*: ISP15-win → DA60/ID15-PRE-blackout (intraday reform); DA60/ID15-POST → DA15/ID15 (DA reform).
  - Same-cal-month + cross-country placebo + bootstrap CIs.
- Honest record of **14 dead causal-identification attempts** (DiD, wind-IV, profile-matching) — rejected after placebo tests.

Difficulty 2 — a structural break in Q1 2024 we cannot yet explain

- Big-4 **nuclear bilateral-contract share** stable at 80–95% across 2018–2023.
- Drops sharply in **Feb–Mar 2024**, three months *before* any MTU15 reform.
- Hydro shows the same Q1-2024 break (auction-side growth); CCGT unaffected.
- Concurrent with the **end of the Iberian gas-cap (Dec 2023)** and the first major renewable-saturation event in Spain (negative DA prices appear Apr 2024).
- Regression on negative-price-share (exogenous proxy, with unit/cal-month FE): nuclear $\beta = -3.05$ ($R^2 = 0.13$); the implied magnitude matches the observed drop. Hydro $\beta \approx 0$.

Mechanism unidentified; reframes how I read 2024 pre-reform data.



What is left

Empirical:

- Pin down the Q1 2024 mechanism (gas-cap / renewables / PPAs).
- Robustness re-pass on remaining alive findings (per-firm Lerner, bid complexification, post-blackout CCGT/aFRR).
- Brodersen-style structural-time-series counterfactual on headline outcomes.
- Per-segment settlement verification once the ESIOS API token clears.

Theoretical & writing:

- Refine the Ito–Reguant extension to motivate the three-lever framing.
- Light HP-style FOC exercise on Big-4 bid functions; no full structural estimation.
- Drafting starts the week after this talk; deadline mid-June 2026.
- **Risk**: 8-week timeline is tight; firm-level Part II is the binding constraint.

Thank you.

Questions and feedback welcome.

`Pablo Paramio -- CEMFI -- May 2026`